DEEP LEARNING: ADVANCED MODELS AND METHODS · DL26

Query-Conditioned Visual-Token Reconstruction

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github.com/SimoneAvellino/VisualTokenRetrieval

PROBLEM

We want to process video online

Process frame-by-frame and discard past frames

What if we need informations about discarded frames?

We use the query to retrieve the past informations.

Memory problems

Keep a long video in memory is not possible. The cost of processing is quadratic.

DATASET

Ego4D: Retrieving Exact Moments in Hours of Egocentric Video

Natural Language Queries (NLQ) Task

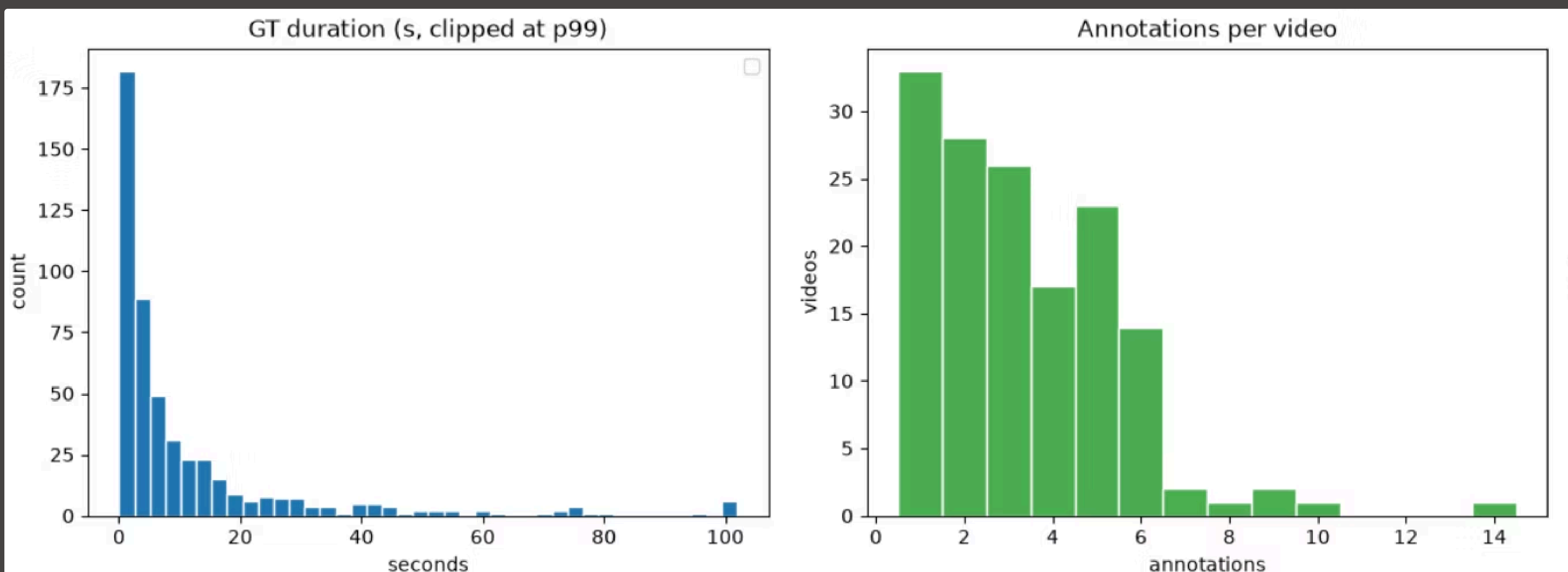
First-person daily-life videos. Each sample links a video, a natural-language question, and a ground-truth time window.

> **Example:** "When did I pick up the cup?" → 45.2–62.5 s

Strict Video-Level Split

- **348** samples for training
- **68** samples for validation
- **44** samples for testing

⚠️ Only 148 videos carried extracted tokens ⚠️



ARCHITECTURE

Baselines

Mean tokens

Returns the mean token of the entire video

Center window

Returns the centered window of the context

EVALUATION

Measuring Reconstruction & Retrieval

Reconstruction metrics:

cos_token

Token-by-token average cosine similarity.

norm_ratio

Predicted/true norm ratio. Ideal = 1.0. Detects scale collapse.

Retrieval metrics:

Recall@K

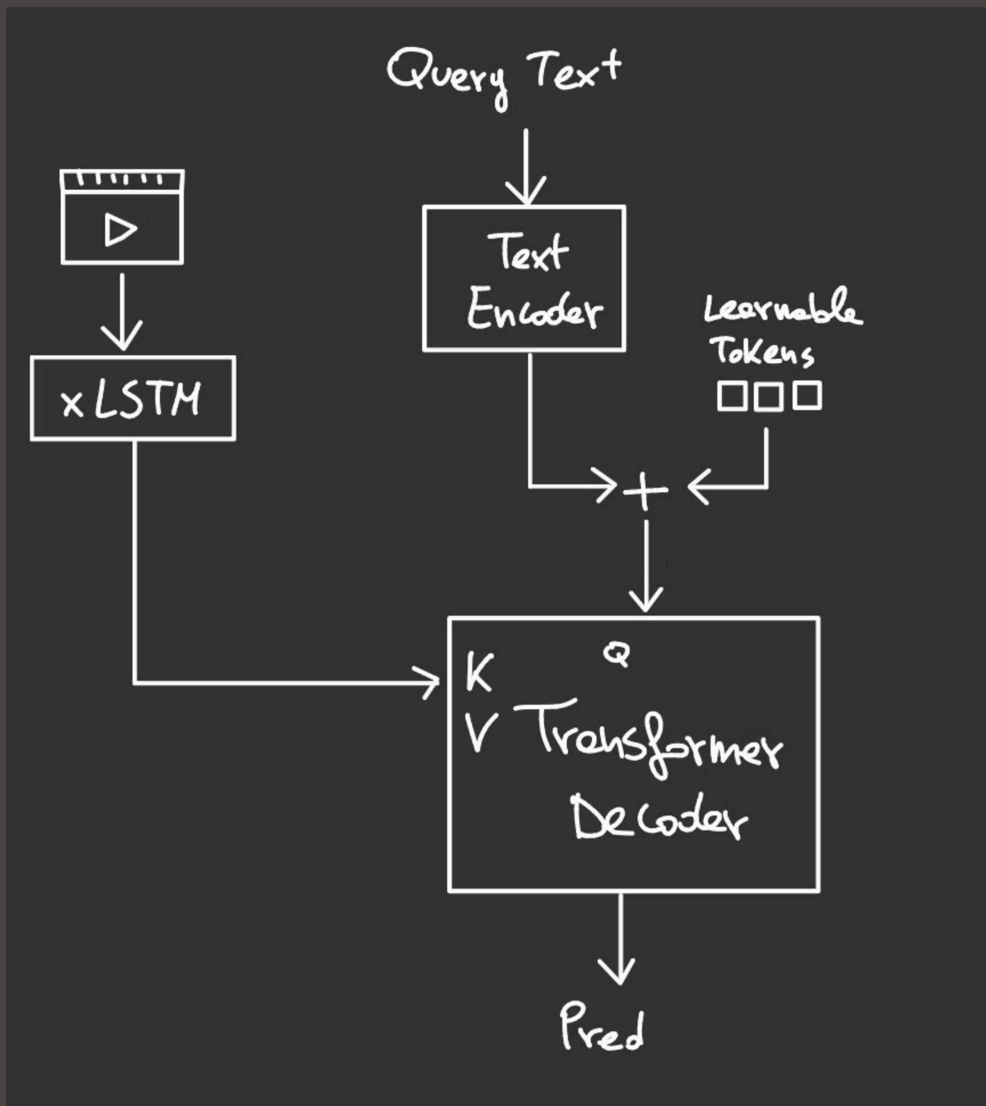
Fraction of predictions where the own GT moment ranks within the top-K most similar candidates.

MRR

Average of 1/rank of the true moment. Higher = the right moment sits near the top.

ARCHITECTURE

Mean Token Reconstructor

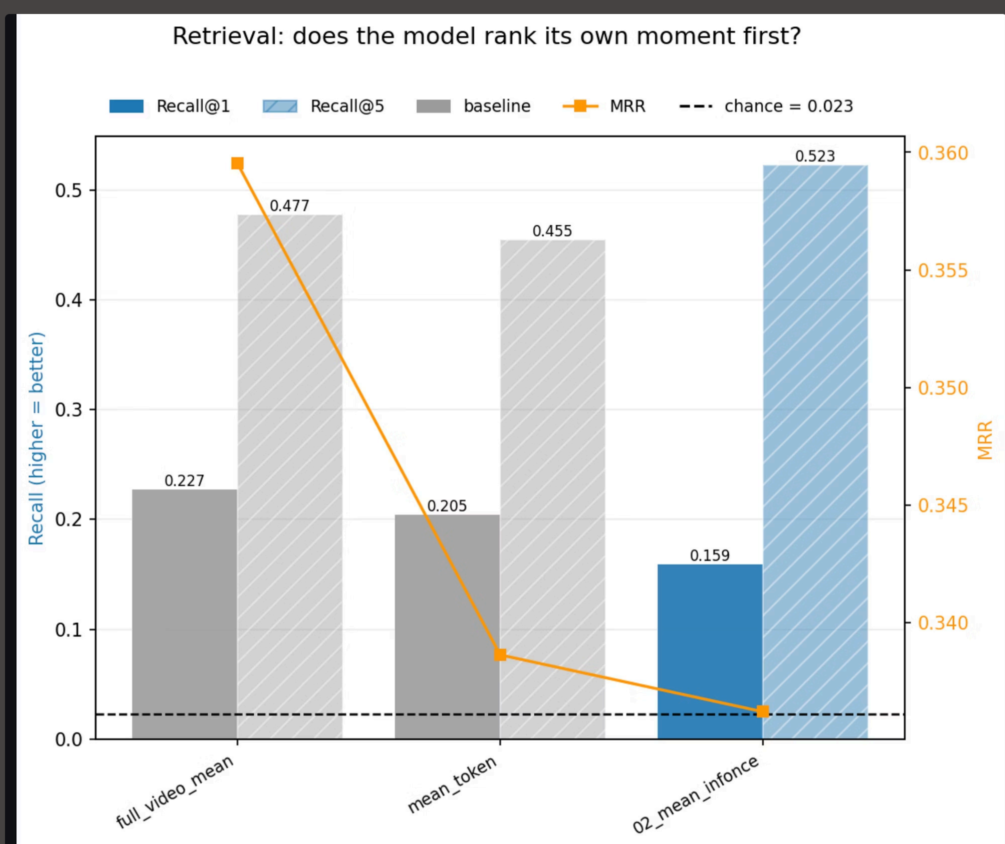


- Video processed by xLSTM → full temporal sequence preserved
- Question (frozen CLIP) guides *where* to look via cross-attention
- Predicts a **single vector** ≈ mean of GT window tokens
- Loss: InfoNCE + cosine anchor

Ablation Study:

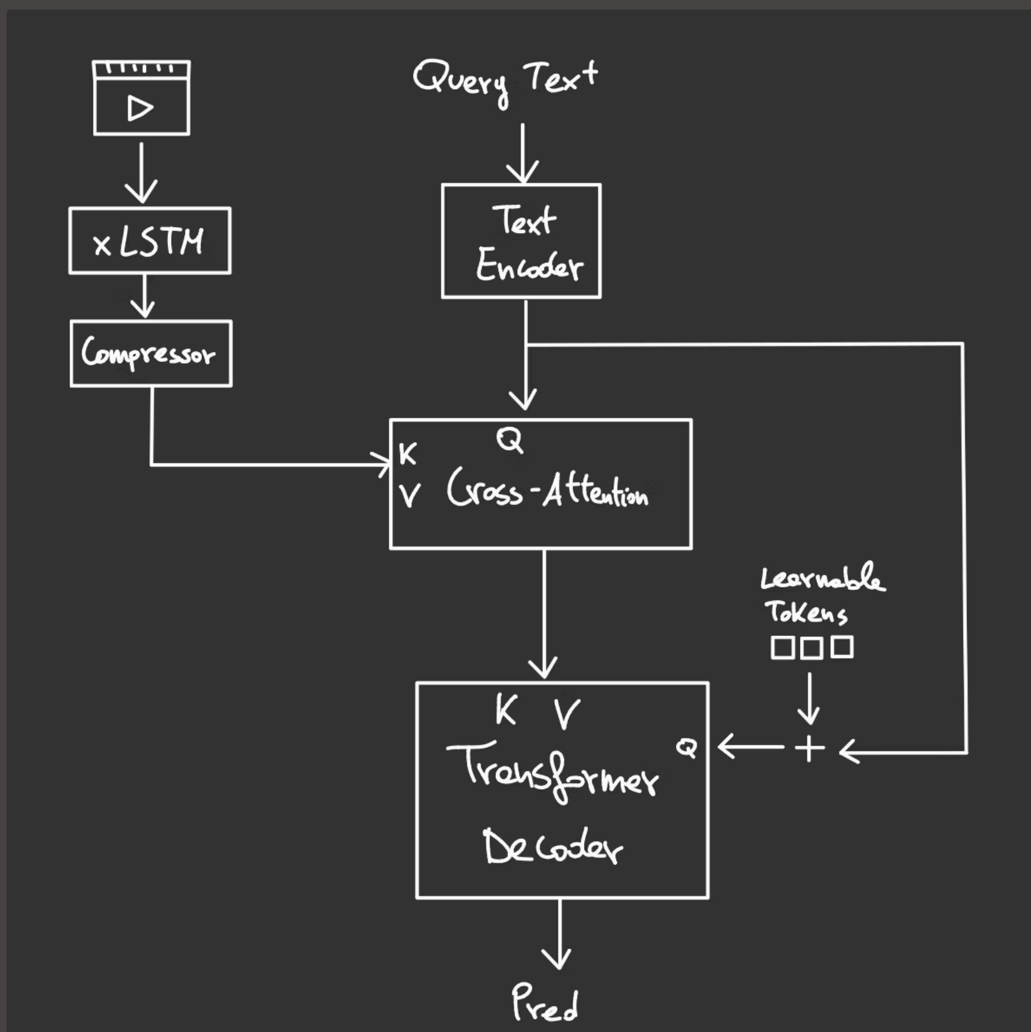
- re-run retrieval replacing each sample's question with a *wrong* question

Right query	Wrong query
• Recall@1: **0.1591** • Recall@5: **0.5227** • MRR: **0.3362**	• Recall@1: **0.1591** • Recall@5: 0.4773 • MRR: 0.3124



ARCHITECTURE

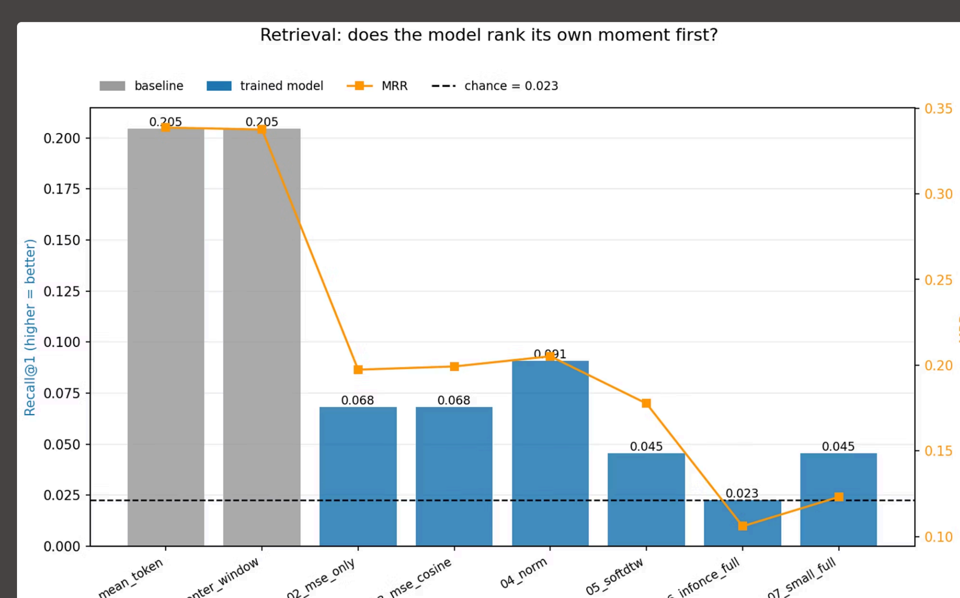
Query-Conditioned Token Reconstructor



- Video compressed by xLSTM + bottleneck → fixed query set Z
- Text tokens cross-attend Z → query-conditioned memory C
- Learnable slots + question summary decode C → **token sequence**

Results on reconstruction 😐

Results on the retrieval 😕



RESULTS

Conclusions

- Mean Reconstructor **ignores the query**
- Trained models do not **surpass the baseline**.
- Query-Conditioned Reconstructor learns **good reconstruction** but **retrieval collapses**
- Train on larger dataset
- Improve how in the InfoNCE negatives are retrieved